

LA CROSSE, WI, USA

WMO: 726430

Lat: 43.879N

Lon: 91.253W

Elev: 652

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14920

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.2	(c) -4.5	(d) -17.6	(e) 2.2	(f) -7.8	(g) -13.2	(h) 2.8	(i) -3.8	(j) 25.3	(k) 19.2	(l) 22.8	(m) 18.3	(n) 5.8	(o) 340	(p) 0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	91.4	75.0	88.6	73.0	85.9	71.5	77.7	87.6	75.7	85.1	73.9	82.9	11.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.7	133.3	83.8	72.6	123.7	81.5	70.9	116.7	79.7	41.6	87.2	39.6	85.5	37.9	83.1	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	19.5	17.8	-16.3	97.2	7.7	4.0	-21.8	100.1	-26.3	102.4	-30.6	104.6	-36.2	107.5
			-16.8	80.9	7.5	2.3	-22.1	82.5	-26.5	83.9	-30.7	85.2	-36.1	86.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	18.2	22.4	35.0	48.5	60.2	70.2	74.5	72.0	64.7	51.4	37.5	24.4
	DBStd	21.78	12.83	12.40	11.98	9.69	8.39	6.72	5.66	5.58	8.06	9.05	9.96	11.60
	HDD50	3676	985	773	485	141	12	0	0	0	3	91	391	795
	HDD65	6929	1450	1193	930	500	197	28	4	8	103	431	826	1259
	CDD50	3089	0	0	21	96	329	606	758	683	445	135	15	1
	CDD65	866	0	0	1	5	49	184	297	225	95	10	0	0
	CDH74	7665	0	0	9	75	497	1624	2753	1839	778	90	0	0
	CDH80	2612	0	0	0	17	146	563	1056	581	232	17	0	0
(14) Wind	WSAvg	8.4	8.5	8.7	8.8	9.6	8.7	7.8	7.5	7.0	7.9	8.8	9.1	8.5
(15) Precipitation	PrecAvg	30.0	1.0	0.9	1.9	3.0	3.8	4.1	3.5	3.6	3.8	2.1	1.5	1.1
	PrecMax	44.1	2.9	2.2	3.7	7.3	9.7	9.5	9.2	9.8	10.5	4.4	3.5	2.6
	PrecMin	17.5	0.2	0.0	0.3	0.6	0.9	1.3	0.2	0.6	0.6	0.2	0.0	0.3
	PrecStd	6.1	0.7	0.6	0.9	1.7	2.1	2.2	2.0	2.4	2.4	1.2	1.0	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	55.3	73.1	82.3	89.9	94.3	97.3	93.1	90.5	82.3	68.3	55.3
		MCWB	41.2	48.6	60.5	63.0	70.5	75.1	78.1	76.9	72.8	67.4	57.1	50.8
	2%	DB	41.6	48.4	64.5	74.5	84.1	89.8	92.2	89.2	86.0	75.8	61.9	46.6
		MCWB	37.1	42.3	54.2	57.4	66.5	73.1	76.4	74.4	71.5	63.0	53.8	41.8
	5%	DB	37.5	42.8	58.3	69.7	79.5	86.2	88.8	85.9	82.2	70.7	56.5	42.4
		MCWB	34.5	37.8	49.9	54.4	64.0	70.7	73.9	72.4	69.0	60.6	49.1	38.6
	10%	DB	34.9	38.5	51.9	64.9	75.2	82.7	85.8	82.9	78.5	66.0	52.0	38.7
		MCWB	32.6	34.8	44.8	51.5	61.7	69.0	72.2	70.6	66.8	57.6	45.6	35.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.6	48.4	62.5	65.1	73.6	77.9	81.5	79.6	76.0	70.7	58.9	51.8
		MCDB	46.7	55.3	71.4	78.7	86.4	89.7	91.4	89.5	86.5	79.0	65.6	54.3
	2%	WB	37.5	42.5	57.0	59.9	69.4	75.4	78.5	76.8	73.2	65.8	54.8	42.8
		MCDB	40.8	47.8	62.5	71.0	79.7	86.3	88.7	85.1	82.1	72.9	60.9	45.1
	5%	WB	34.9	38.3	50.4	56.4	66.7	73.3	76.3	74.8	71.1	62.0	50.1	38.6
		MCDB	37.6	42.6	57.7	66.5	75.7	83.6	86.0	82.8	79.1	69.0	55.7	42.0
	10%	WB	32.7	35.0	44.6	53.0	63.8	70.9	74.3	72.9	68.8	58.4	46.1	35.5
		MCDB	34.9	38.2	50.9	63.3	72.7	79.8	83.2	80.6	76.4	65.2	51.5	38.3
(35) Mean Daily Temperature Range	MDBR		15.3	16.5	17.8	19.6	20.2	19.3	19.4	18.9	19.9	17.9	14.7	13.3
	5% DB	MCDBR	16.4	18.1	24.8	27.6	25.5	22.9	22.1	21.3	23.3	23.9	21.3	16.2
		MCWBR	13.1	13.1	14.8	14.4	11.9	9.9	9.3	9.0	10.8	13.4	14.2	12.8
	5% WB	MCDBR	15.8	17.6	22.9	23.5	22.0	20.3	19.7	18.7	20.0	20.7	20.1	14.6
		MCWBR	13.0	13.2	14.5	14.0	11.6	10.4	9.7	9.4	10.7	13.7	14.8	12.2
(40) Clear-Sky Solar Irradiance	taub		0.266	0.296	0.327	0.367	0.397	0.408	0.412	0.406	0.373	0.332	0.297	0.269
	taud		2.391	2.314	2.375	2.326	2.294	2.297	2.291	2.318	2.392	2.496	2.511	2.444
	Ebn at Noon		277	285	289	284	278	274	271	269	270	269	264	265
	Edh at Noon		24	31	34	38	41	41	41	38	33	26	22	21
(44) All-Sky Solar Radiation	RadAvg		565	904	1230	1436	1700	1888	1955	1659	1333	855	560	413
	RadStd		58	88	143	138	130	146	91	112	99	113	48	42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page